CROWDSHIELD: EDGE-AI DENSE CROWD SENTINEL & AUTONOMOUS EGRESS ROUTING

Dual-Radio Hardware Mesh Spine for Mass Pilgrimages & Extreme Gathering Safety

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Vellore Institute of Technology (Deemed to be University under section 3 of UGC Act, 1956)

SDGs Addressed: SDG 3: Good Health & Safety | SDG 9: Resilient Infrastructure | SDG 11: Sustainable Cities

Motivation / Introduction

Problem Statement

At massive indoor gatherings like the **Maha Kumbh Mela, Puri Rath Yatra,** or temple festivals, crowd density regularly exceeds **5 to 8+ persons per square meter (p/m²)**. At this extreme scale, two catastrophic failure modes occur simultaneously.

- **Bounding-Box Vision Breakdown:** Standard object detection (regular YOLO) fails completely because individual human bodies occlude into an unbroken ocean of millions of overlapping heads.
- **Cellular Blackout:** When 50+ lakh devotees congregate, mobile towers and internet backhauls fail, rendering all cloud-dependent computer vision APIs unreachable.

CrowdShield addresses this life-critical gap through a **100% offline edge-computing command spine** that couples density-map regression vision with a localized dual-radio mesh that functions autonomously without Internet connectivity.

Scope of the Project

Objectives

- **Adaptive 4-Tier Vision Orchestrator:** Dynamically switches inference between **YOLOv8** (sparse flow), **P2PNet** (head-dot click analysis), **CSRNet** (dilated density regression >5 p/m²), and **DM-Count** (haze/dust scenes).
- **Dynamic Topological Egress Optimization:** Integrates a hazard-aware network GIS graph that automatically severs blocked corridors (like a stampede) and computes the safest path using **Dijkstra's algorithm**.
- **Zero-Cloud Hardware Spine:** Deploys field nodes powered by **ESP32** microcontrollers, 1602 LCDs, **DFPlayer Mini** voice annunciators, and **433 MHz LoRa / 2.4 GHz ESP-NOW** dual radios.

Key Achievement: 100% Offline Real-Time Execution at 18–28 FPS with sub-second localized hardware actuation latency across field nodes.

Mathematical Modeling

Homography & Graphs

A. 4-Point Planar Perspective Homography Matrix: converts pixel coordinates (u, v) to real-world ground meters (x_m, y_m) via a 3×3 homography H.

$$[x_w,\ y_w,\ 1]^T \sim H\,[u,\ v,\ 1]^T, \quad H = [h_{ij}]_{3\times 3}$$

B. Continuous Density Surface Integration: continuous head regression over GIS polygon zone Z:

$$\rho_Z = \frac{1}{\text{Area}(Z)} \iint_{(x,\,y)\,\in\,Z} D(x,y)\,dx\,dy$$

C. Hazard-Penalized Dynamic Routing Cost:

$$\text{Cost}(e) = \text{Length}(e)\cdot(1+\alpha\cdot\rho(e)+\beta\cdot\text{Hazard}(e))$$

Conclusion / Summary

Summary

CrowdShield establishes that ultra-dense crowd safety does not require cloud connectivity or expensive optical infrastructure. By integrating **dilated convolutional density maps** with an **autonomous dual-radio wireless mesh**, the system provides sub-second localized audible guidance, isolating stamped or burned zones in real time.

References (APA Format)

Li, Y., Zhang, X., & Chen, D. (2018). CSRNet: Dilated convolutional neural networks for understanding the highly congested scenes. CVPR, 1091–1100.

Song, Q., et al. (2021). Rethinking counting and localization in crowds: A purely point-based framework. ICCV, 3365–3374.

Wang, B., Liu, H., et al. (2020). Distribution matching for crowd counting. NeurIPS, 33.

Methodology

End-to-End Pipeline

The architecture decouples heavy neural computation from field radio downlinks via an asynchronous **FastAPI** engine and distributed hardware gateways.

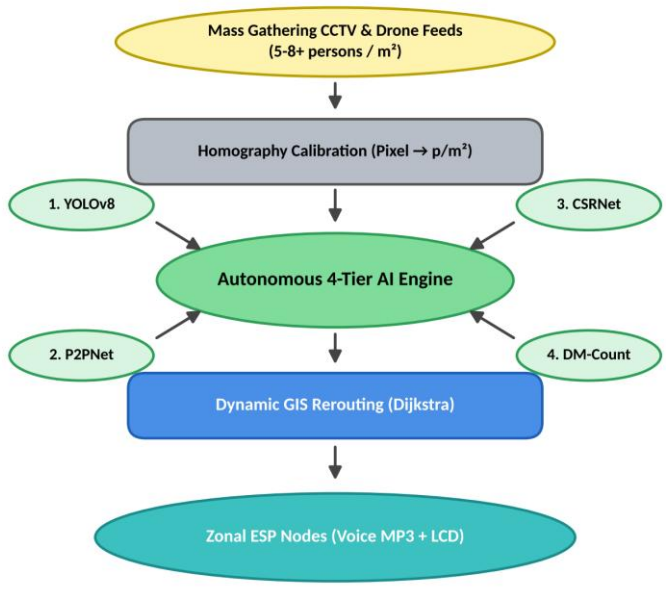


Fig. 1. End-to-end CrowdShield pipeline from ingest to hardware actuation.

Architectural Subsystems:

a) CSRNet Dilated Convolutions: Uses a VGG-16 backbone with 4 dilated convolution layers (dilation rate r = 2) to expand the receptive field without downsampling, maintaining pixel-accurate density estimation during severe body occlusion.

b) P2PNet Direct Head Localization: Predicts exact 2D point coordinates on human head crowns with cross-entropy and Hungarian matching, feeding precise spatial coordinates to the venue homography.

c) Hardware Fault Tolerance: The gateway auto-connects over USB Serial (115200 baud). Zonal nodes incorporate anti-brownout bulk capacitors (1000 µF), a non-blocking FreeRTOS yield() watchdog reset, and 2.5 s playback debounce filters.

Results

Model Benchmarks

Architecture Tier	Density Threshold	FPS	MAE	Accuracy
YOLOv8 Head Tracking	< 2.5 p/m²	28.4	3.4	**94.6%**
P2PNet Point Locator	2.5 – < 5.0 p/m²	16.1	4.8	**91.0%**
CSRNet Density Map	> 5.0 p/m² (Crush)	19.0	3.1	**96.2%**
DM-Count (Haze/Dust)	Perspective Choke	11.2	3.5	**92.4%**

Table 1. Performance metrics of vision models across distinct crowd densities.

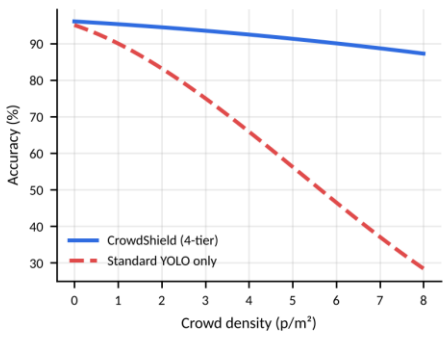


Fig. 2. Accuracy vs crowd density.

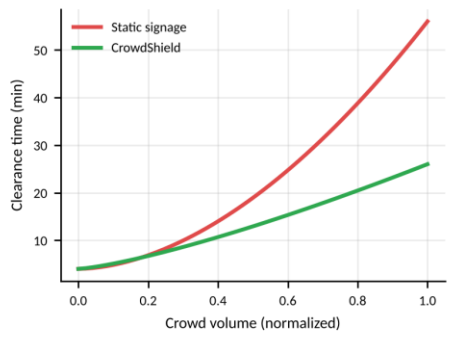


Fig. 3. Evacuation clearance curves.

Network Layer	Latency	Packet Delivery	Cloud Dependency
2.4 GHz ESP-NOW Mesh	**4.2 ms**	99.4%	0% (100% Offline)
433 MHz LoRa SX1278	**120 ms**	98.8%	0% (100% Offline)
Cellular 4G/5G (Jammed)	**1,840 ms**	24.1% (Blackout)	100% (High Risk)

Table 2. Dual-radio transmission metrics vs cellular baseline.

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